Aaryan Wadhawan

+91-8130785343 | aaryanwadhawanedu@gmail.com | linkedin.com/in/aaryan-wadhawan1410 | github.com/aaryanwadhawan7

Education

VIT Bhopal University **2022 – June 2026 (Graduated)**

B.Tech in Computer Science (Cyber Security and Digital Forensics), CGPA: 8.15/10.0

Projects

Autonomous Coding Agent | GitHub

- Built an autonomous **Python** coding agent using **Groq function calling** to execute generated code, analyze stdout/stderr, and iteratively self-correct errors for up to 5 iterations.
- Engineered an isolated **Docker sandbox** with a 128MB memory limit, disabled networking, and a 5-second execution timeout for safe execution of untrusted code without persistent container state.
- Developed an asynchronous **FastAPI** backend using thread-pool execution for non-blocking Docker and LLM calls, with a **Streamlit** UI featuring live iteration tracking and light/dark mode.

DocuMind – RAG Document Q&A System | GitHub

- Built an end-to-end **RAG pipeline** using FastAPI, LangChain, FAISS, and LLaMA 3 (Groq API); users upload PDFs and receive context-grounded answers with source citations at **sub-2-second latency**.
- Engineered document ingestion with **500-token chunking** and semantic vector indexing; retrieves top-3 relevant chunks via cosine similarity before passing context to the LLM.
- Integrated **MLflow** experiment tracking and **Prometheus + Grafana** monitoring; logs every inference run with latency and model metadata for performance comparison.
- Containerised full stack with **Docker Compose** (5 services); **GitHub Actions CI/CD** runs pytest on every push and builds Docker image only on passing tests.

ARistotle – AI-Powered AR Object Explainer | GitHub

- Built a browser-native AR object explainer using **YOLOv8s** for real-time object detection on live webcam frames every 3.5s; detected labels passed to **LLaMA 3 via LangChain** to generate explanations rendered as floating AR cards using AR.js and A-Frame, with no app install required.
- Implemented in-memory **LLM response caching** that cut Groq API calls by ∼90% per session, improving efficiency and keeping inference within free-tier limits at sub-500ms response time.
- Optimized memory footprint on a GPU-less Render server by switching to **CPU-only PyTorch wheels**, resolving a silent OOM crash from CUDA bindings; deployed live with FastAPI backend and Netlify frontend.

Technical Skills

Languages: Python, C++, C, SQL, JavaScript
AI Engineering: LLMs, Agentic AI, Function Calling, RAG, LangChain, FAISS, Generative AI, Inference Optimization
Machine Learning & CV: Machine Learning, Deep Learning, Neural Networks, PyTorch, YOLOv8, AR.js, A-Frame
MLOps & DevOps: Docker, Docker Compose, GitHub Actions, pytest, Prometheus, Grafana, MLflow
Backend & Web: FastAPI, REST APIs, Streamlit, HTML, CSS
Databases: MongoDB, SQL
Cloud & Tools: Render, Netlify, Git, GitHub

Achievements

- Ranked in **Top 50** at Visionary Hackathon 2.0 (Metti Xathon) with ARistotle.
- Solved **250+ DSA problems** across LeetCode, GeeksforGeeks, and Codeforces.

Certifications

- Oracle Generative AI Professional
- IBM DevOps, Agile and Design Thinking